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FindMine: Detecting Personal Objects and Identifying Their Owners in Real-World Images

NHL Stenden Professorship in Computer Vision & Artificial Intelligence

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1 Introduction

Finding an object in a shared environment requires more than recognizing its category. A detector can locate a wallet or travel mug, but it does not normally determine which registered person owns that particular instance. Ownership recognition is challenging because objects belonging to different people often have the same shape and differ only in colour, texture, branding, or printed text.

A two-stage design is therefore proposed: the first stage detects the category and location, and the second classifies the cropped object by owner or rejects it as *Unknown*. Object detection was selected instead of segmentation because rectangular regions are sufficient for extracting classifier inputs, while requiring substantially simpler annotations. YOLO provides an efficient single-stage detection formulation [1]; this study uses Ultralytics YOLO11 [2].

Related work. Object-instance re-identification has combined foreground and background cues in fixed indoor scenes [3]. Recent instance-detection work uses multi-view references and pretrained visual features to match specific objects in novel scenes [4, 5]. FindMine instead evaluates a supervised category-then-owner pipeline for registered belongings; results on different datasets are not directly comparable.

The research question is: *How accurately can a YOLO11s detector combined with transfer-learned classifiers identify both the category and owner of wallets and travel mugs in real-world images?*

2 Methods

Data collection. The detection dataset contained 280 images: 267 photographs collected for this project and 13 externally sourced wallet images. The project photographs spanned five batches and targeted hard cases. Images varied in background, viewpoint, scale, occlusion, lighting, number of target objects, and presence of negative scenes.

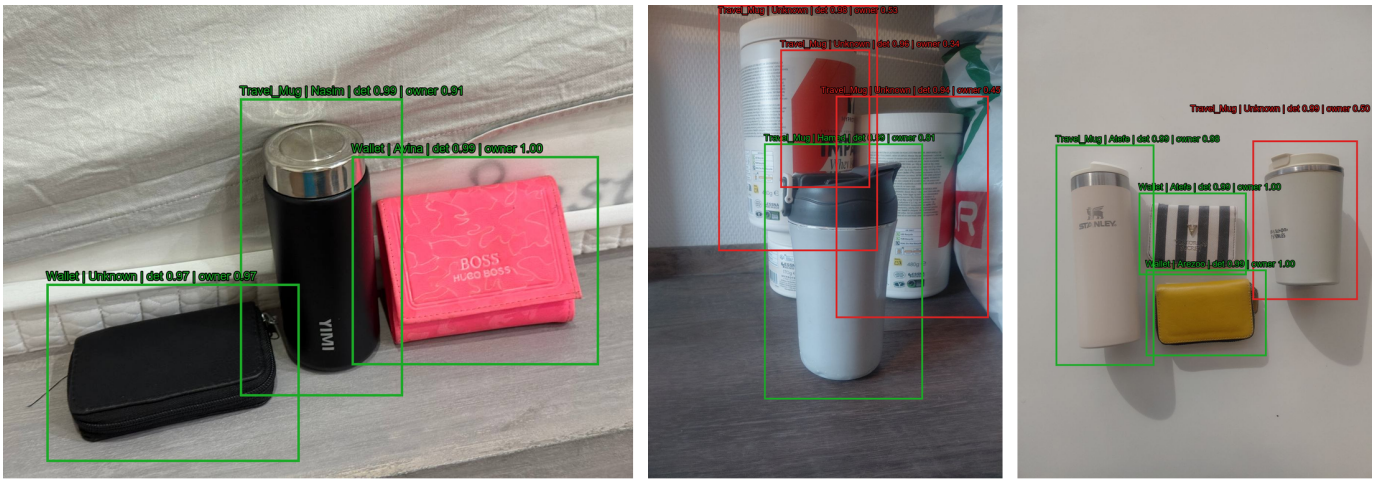
Annotation and grouped split. Bounding boxes were annotated for *Wallet* and *Travel_Mug*. To reduce leakage from near-duplicate views, images were grouped by scene identifier before splitting into 208 training, 40 validation, and 32 test images. The corresponding object counts were 244, 52, and 50; the splits contained 26, 1, and 1 background-only images, respectively. The training objects comprised 127 wallets and 117 travel mugs, while the test set contained 19 and 31 instances. The 49 training-only support images comprised 36 project photographs and 13 externally sourced images; detector validation and test used only the five photographed batches. Early pilot predictions exposed missed wallets and false positives on background objects. We corrected labels, added hard and negative scenes, and expanded the training data before the final grouped split. YOLO training used mosaic, HSV colour jitter, random scaling and translation, and horizontal flips; MixUp and rotation were disabled.

Models and training. A COCO-pretrained [6] YOLO11s detector was fine-tuned at 640×640 resolution with batch size 16 for up to 100 epochs (early-stopping patience 20). Ultralytics' automatic optimizer selection chose AdamW; validation selected the checkpoint. Its boxes provide the crops used by the ownership stage. For owner recognition, ground-truth boxes were cropped to prevent detector errors from influencing classifier comparison. The travel-mug dataset contained 156/40/41 training/validation/test crops from eight registered owners plus *Unknown*; the wallet dataset contained 166/34/29 crops from six owners plus *Unknown*. External *Unknown* crops from a wallet dataset [7] and a tumbler collection [8] entered classifier training, validation, and test. Splits were owner-aware and grouped so closely related views remained together.

ImageNet-pretrained [9] ResNet18 [10] and MobileNetV3-Small [11] were trained on 224×224 crops for 35 epochs with batch size 16 and class-weighted cross-entropy. Training crops used random resized cropping, horizontal flipping, rotation up to 10°, and colour jitter. ResNet18 uses residual skip connections; MobileNetV3-Small served as a lightweight comparison. AdamW trained the classification head for five epochs with the backbone frozen (learning rate 10^{-3}), then fine-tuned the full network (learning rate 10^{-4}; weight decay 10^{-4}), following transfer-learning principles [12]. Both models assigned each crop to a registered owner or *Unknown*.

Evaluation. We report detection precision, recall, F1, mAP@0.5, and mAP@0.5:0.95, alongside owner accuracy and macro-F1. For end-to-end testing, detector crops entered the selected ResNet18. Detection confidence 0.88 was selected on validation; matches required IoU ≥ 0.5, and owner predictions below confidence 0.50 were rejected.

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(a) Three correct predictions (b) Three false positives (c) One owner error

Fig. 1. Full-pipeline examples. Green boxes mark correct category and owner predictions, including a correctly rejected *Unknown*; red boxes mark false-positive detections or owner errors. (a) Three correct objects. (b) One correct mug and three mug-like false positives. (c) Three correct objects and Arezoo's mug misclassified as *Unknown*. Labels show detection and owner confidence.

3 Results

On the grouped detector test set, YOLO11s obtained class-averaged 0.840 precision and 0.696 recall in the standard YOLO summary (at its F1-optimal confidence), alongside 0.828 mAP@0.5 and 0.752 mAP@0.5:0.95. Wallet and travel-mug mAP@0.5 values were 0.890 and 0.765, respectively. Table 1 summarizes owner classification. ResNet18 achieved the strongest validation macro-F1 for both object types, while MobileNetV3-Small achieved perfect accuracy on the small wallet test set.

Table 1. Owner-classification results. Model selection uses validation macro-F1.

Object	Model	Val. F1	Test acc.	Test F1
Travel mug	ResNet18	**0.887**	**0.927**	**0.943**
Travel mug	MobileNetV3-S	0.705	0.854	0.815
Wallet	ResNet18	**1.000**	0.966	0.984
Wallet	MobileNetV3-S	0.968	**1.000**	**1.000**

Separately, at the validation-selected pipeline threshold of 0.88, 39 of 50 objects were matched; 11 were missed and seven detections were false positives. Micro-averaged detection precision, recall, and F1 over these objects were 0.848, 0.780, and 0.813; conditional owner accuracy was 0.923. Counting misses, false positives, and owner errors, end-to-end precision, recall, and F1 were 0.783, 0.720, and 0.750. Ownership was correct for all 14 matched wallets and 22 of 25 matched mugs (Fig. 2). Wallets had five misses and one false positive; mugs had six misses, three owner errors, and six false positives.

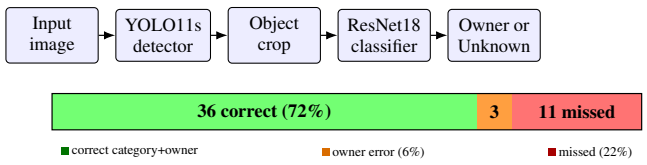


Fig. 2. FindMine processing stages and end-to-end outcomes for 50 ground-truth test objects. Seven additional detections were false positives.

4 Discussion and Conclusion

The results answer the research question positively but conditionally: on the present grouped test set, the pipeline correctly completed detection and ownership recognition for 72% of all ground-truth objects, with an end-to-end F1 of 0.750. The 0.923 conditional owner accuracy indicates that ownership recognition is usually reliable once an object is correctly localized. The comparison also shows that the larger ResNet18 backbone was more consistent under validation-based selection, although the lightweight MobileNetV3-Small remained competitive.

Detection remained the bottleneck: eleven misses and seven false positives. Early pilots also confused large background regions, including sofas, with target objects. We added hard negatives and tuned the confidence threshold on validation, but did not isolate the role of apparent object scale. Some correct boxes still extended beyond the object, potentially introducing background into owner crops (Fig. 1). Better localization and negative sampling merit further study.

Evaluation is limited by few physical objects, repeated views, small per-owner test counts, and few participants. Supervised *Unknown* is not formal open-set recognition [13]; externally sourced crops may introduce style cues. Exploratory runs changed both data and splits, so gains cannot be assigned to one intervention. These results are a proof of concept, not evidence of broad generalization. Future work should test unfamiliar objects photographed under matched conditions and compare lost-item matching with manual search.

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